Trajectory-conditioned ACT

Causal RGB and sparse 3D track histories for bimanual garment folding

RGB · Point tracks · Training only

a Causal observations

Current RGB



RGB history



16 frames, 3 views, 0.5 s

Current joint state

$q_t \in \mathbb{R}^{12}$

Observed cloth tracks



K = 16 points
T = 16 times
GT 3D positions

Past and present only

Observed tracks:
actor input
Hidden tracks:
auxiliary targets only

$L = L_{action} + 10\,L_{KL} + \lambda\,L_{comp}$

b Observation encoding

ResNet-18
Shared weights
ImageNet init.



Full spatial grid
3 × 15 × 20 tokens

4 × 4 pooling
15 × 3 × 16 tokens

512-D projection + position / time codes

Joint projection
Normalize, 12 → 512

Shared trajectory MLP
One 512-D token per observed point

All observation tokens have width 512

c Native ACT policy

CVAE latent
Train: posterior from q_t and GT future actions
Evaluate: z = 0 latent width 32

Concatenate

Observation
tokens O_t



ACT encoder
4 layers
8 heads
d = 512

ACT decoder
1 layer
8 heads
d = 512

Action queries
100 learned positions

Pre-Transformer tap
Latent excluded

Native actor, CVAE and action loss retained

d Action chunk

Action head
512 → 12
Inverse normalization



$\hat{A}_t \in \mathbb{R}^{100\times12}$

100 future joint targets
12 output dimensions

Validation: raw action MSE
Denormalized targets, MSE in rad²

e Masked track completion

TRAINING ONLY

Hidden-point queries
48 current XYZ + 15 past times

Memory projection
O_t : 512 → 128

Completion decoder
1 layer, 4 heads, d = 128

Past displacements
48 × 15 × 3

Completion loss
λ · Smooth L1

Query MLP: [XYZ, time] → 128

GT hidden past motion

Blue / teal: actor inputs Amber: training-only computation

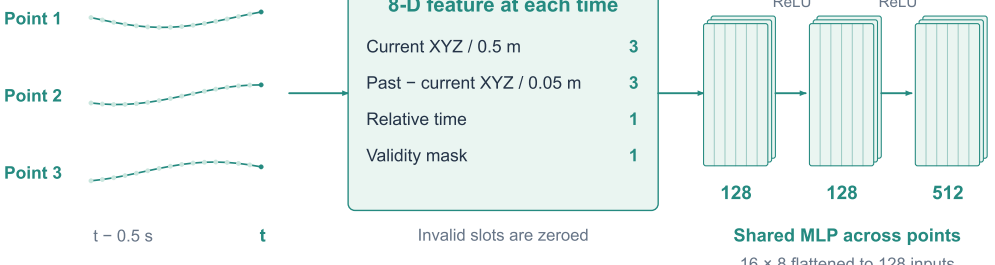
Auxiliary gradients update observation encoders / projections, not the main ACT Transformer directly

Trajectory encoding and experimental controls

Implemented design: fixed causal windows, shared ACT core and an auxiliary-gradient control

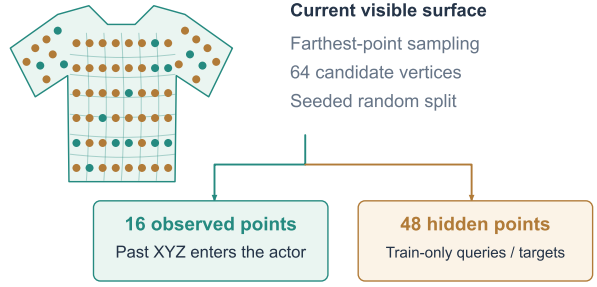
a Point-wise trajectory representation

Persistent point identities within each window



K = 16 trajectories, T = 16 slots

b Observed / hidden point split



Resample identities across windows, retain identity across time

c Matched observation controls

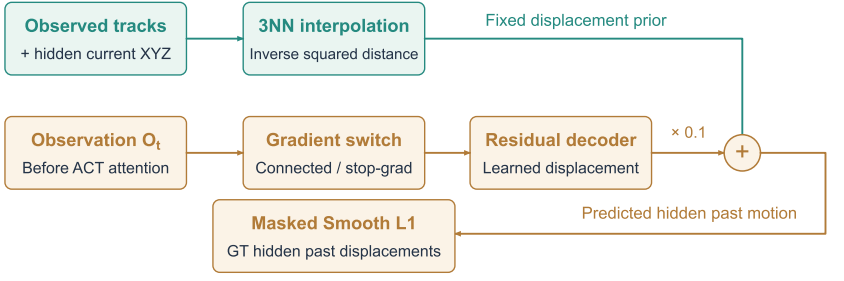
Every condition uses current RGB and current joint state

Condition	RGB history	Current XYZ	Past motion	Aux. loss
ACT	—	—	—	—
RGB-H	16 frames	—	—	—
Geometry	16 frames	✓	—	—
Tracks	16 frames	✓	✓	—
Tracks + Completion	16 frames	✓	✓	✓

T = 16 at stride 3, 90 Hz. Predict 100 joint targets in every condition.
History-enabled variants share the RGB window. GT point tracks are privileged inputs.

d 3NN residual completion control

Same actor inputs and architecture in both variants



Connected: auxiliary gradients reach shared observation encoders.
Detached: the auxiliary head learns, with its memory input detached.